Eine Woche, ein Beispiel

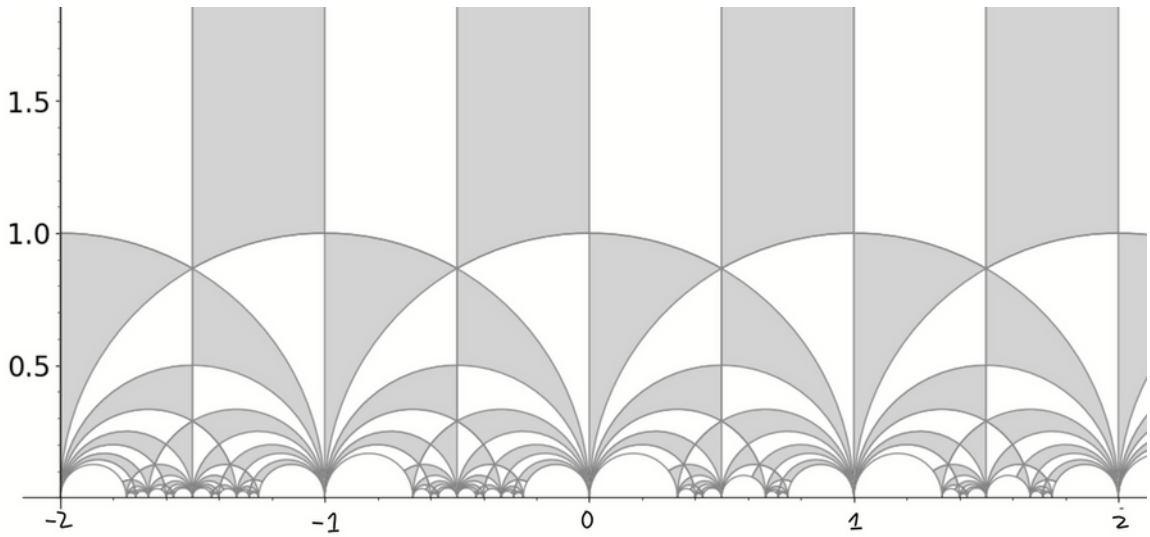
12.28 binary quadratic form

This note should be inserted in [2025.12.21], Section 2.
In fact, the geometry of binary quadratic forms is a bit interesting, regardless of its relation with Hodge structures.

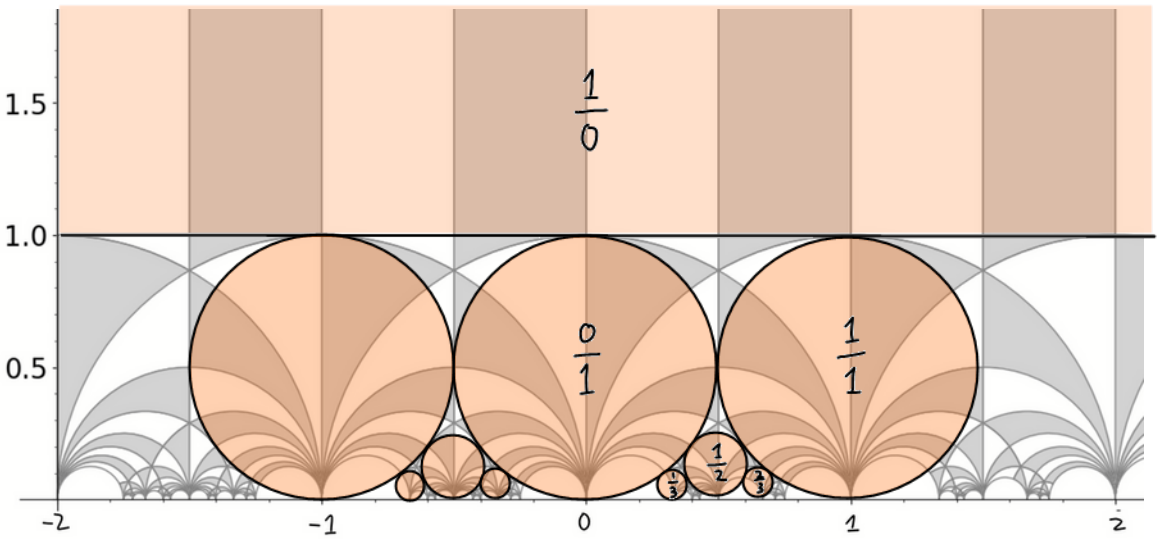
1 several figures in $\mathcal{H}$
2. equivalent expressions of $M_{sym}^{2\times 2}(\mathbb{R})$
3. some invariants

1 several figures in $\mathcal{H}$ [JC97, Chapter 1]

①. Fundamental domain of $SL_2(\mathbb{Z})$



②. Ford circles

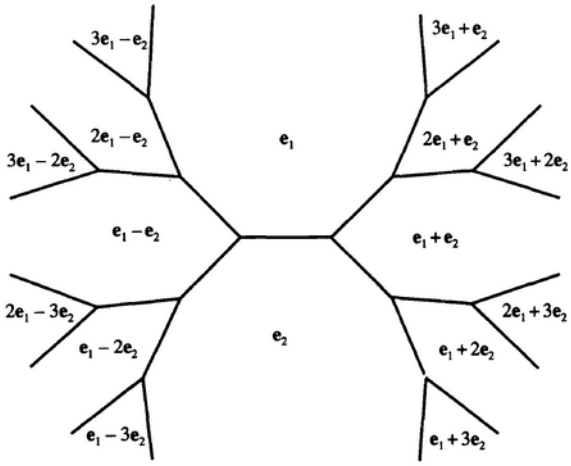


$$\mathbb{Q} \cup \{\infty\} = \text{“}\mathbb{P}\mathbb{Z}^2 = (\mathbb{Z}^2 - \{0\}) / (\mathbb{Z} - \{0\})\text{”} = SL_2(\mathbb{Z}) / \begin{pmatrix} 1 & * \\ 0 & 1 \end{pmatrix}$$

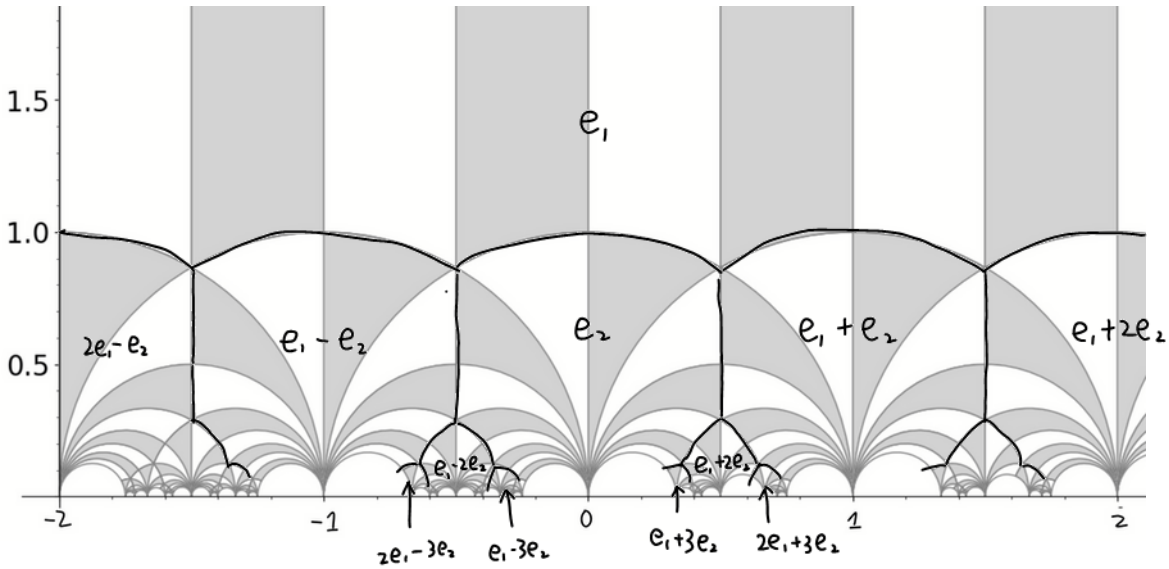
The "distance" between two circles $\frac{p_1}{q_1}$ & $\frac{p_2}{q_2}$ is not $\begin{vmatrix} p_1 & p_2 \\ q_1 & q_2 \end{vmatrix}$

How to compute it?

③ Conway's topograph



[JC97, p8]



$$\mathbb{Z}^2_{prim}/\{\pm 1\} \cong \mathbb{Q} \cup \{\infty\}$$

Ford circle related with SL_2(Z)-action:
https://mathoverflow.net/questions/25676/canonical-fundamental-domain-for-a-discrete-subgroup-%ce%93-of-sl%e2%82%82r-acting-on-hyper
I thought that the afterthoughts in [JC97] is more useful.

perpendicular bisector:
https://math.stackexchange.com/questions/2220933/perpendicular-bisector-of-a-hyperbolic-line
https://www.researchgate.net/publication/221435311_Hyperbolic_Voronoi_Diagram (Lemma 2,3 gives what I want)

geometrical intuition of hyperbolic circles:
https://www.youtube.com/watch?v=mk2et2_2zzk

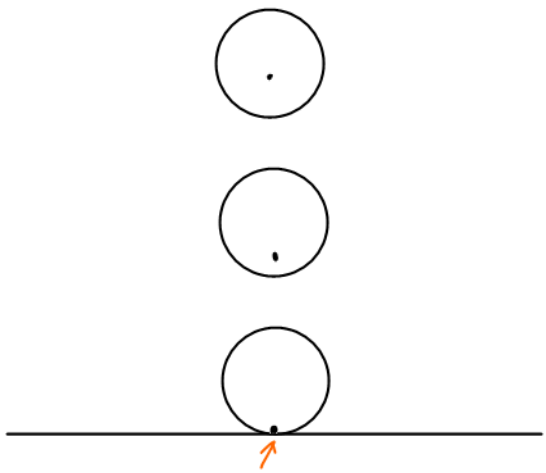
Rmk.

Be careful that the hyperbolic center is not the usual euclidean center.

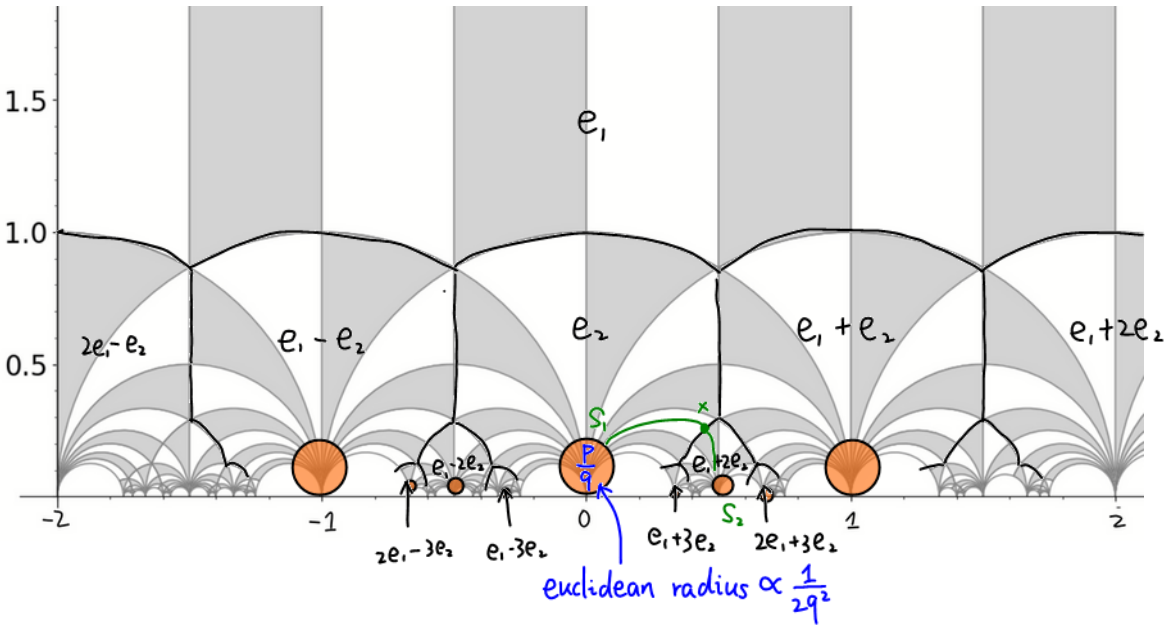
Using the hyperbolic metric,
The Voronoi diagram of the Ford circles gives the boundary of a Ford domain.

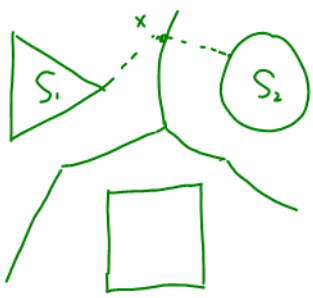
You can also shrink the Ford circles in the same percentage, and the Voronoi diagram is still invariant.

Therefore, You can view the Ford domain as the Voronoi diagram of set of rational numbers.

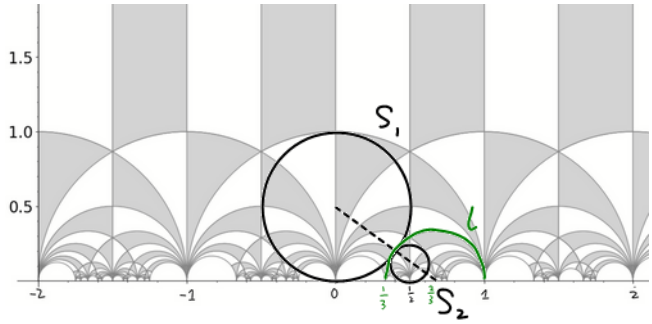


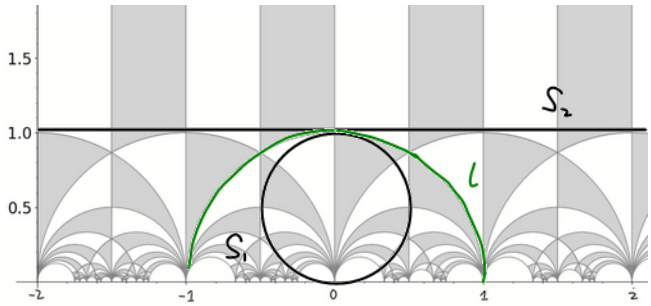
This is the "center", with radius $+\infty$





Voronoi diagram:
$d(x, S_1) = d(x, S_2)$





Observe: In the above cases,

$l = \{x \in \mathcal{H} \mid d(x, S_1) = d(x, S_2)\}$ is a half circle, and the inversion transformation w.r.t. l switches S_1 & S_2.

Therefore, given S_1, S_2, l, and two can uniquely determine the third, and two conditions give the third:

① S_1 has $+\infty$ radius. i.e., meet $\mathbb{R} \cup \{\infty\}$ at 1 pt

② S_2 has $+\infty$ radius

③ l is a half circle i.e., a geodesic

Rmk. 1. Why Conway's topograph is useful:

To understand $\beta(v, w)$, it reduces to understand

$$\beta(v, v) \quad \text{for} \quad v \in \mathbb{Z}^2_{prim}/\{\pm 1\}$$

2. The higher-dim generalization would be $SL_n(\mathbb{Z}) \backslash SL_n(\mathbb{R}) / SO(n)$.

see https://arxiv.org/abs/math/0601084

2. equivalent expressions of $M_{sym}^{2\times 2}(\mathbb{R})$

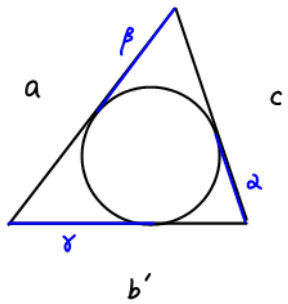
Given a bilinear form $\beta: \mathbb{Z}^2 \times \mathbb{Z}^2 \longrightarrow \mathbb{R}$, we have

$$A_\beta = \begin{pmatrix} a & \frac{b}{2} \\ \frac{b}{2} & c \end{pmatrix} \in M_{sym}^{2\times 2}(\mathbb{R}) \iff q_\beta(x,y) = ax^2 + bxy + cy^2$$

Here, a, b, c don't have equal status. If we set

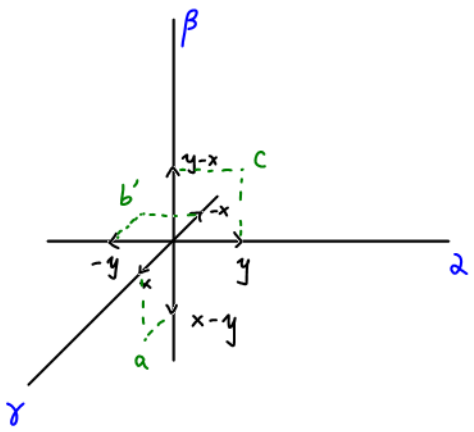
$$\begin{cases} b' = a+b+c, \\ \alpha = \frac{1}{2}(b'+c-a) \\ \beta = \frac{1}{2}(c+a-b') = -\frac{1}{2}b \\ \gamma = \frac{1}{2}(a+b'-c) \end{cases}$$

β is a bilinear form if it doesn't emerge with α, γ in the same time.

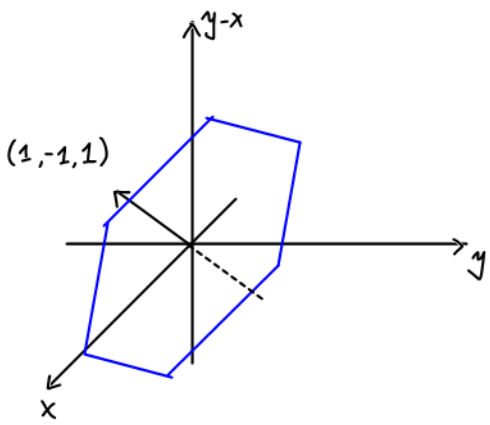


then

$$\begin{aligned} q_\beta(x,y) &= ax^2 + bxy + cy^2 \\ &= ax^2 + (b'-a-c)xy + cy^2 \\ &= ax(x-y) + b'xy + cy(y-x) \\ &= (\beta+\gamma)x^2 - 2\beta xy + (\alpha+\beta)y^2 \\ &= \alpha y^2 + \beta(x-y)^2 + \gamma x^2 \end{aligned}$$



Why symmetry



The plane where q_β can take values on

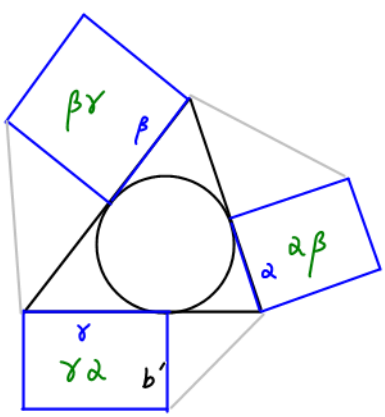
Therefore, every condition/invariants has many equivalent descriptions:

Assume $A_\beta \in M^{2\times 2}_{sym}(\mathbb{Z})$

	integral quadratic form	integral bilinear form	primitive
	$\mathrm{Im}\, q_\beta \in \mathbb{Z}$	$\mathrm{Im}\, \beta \in \mathbb{Z}$	q_β is primitive
A_β	$A_\beta \in M^{2\times 2}_{sym}(\mathbb{Z})$	$A_\beta \in M^{2\times 2}_{sym}(\mathbb{Z})$	
(a,b,c)	$a,b,c \in \mathbb{Z}$	$a, \frac{b}{2}, c \in \mathbb{Z}$	$\gcd(a,b,c)=1$
(a,b',c)	$a,b',c \in \mathbb{Z}$	$a,b',c \in \mathbb{Z},\ a+b'+c \in 2\mathbb{Z}$	$\gcd(a,b',c)=1$
(α,β,γ)	$\alpha,\beta,\gamma \in \mathbb{Z}$ or $\alpha,\beta,\gamma \in \mathbb{Z}+\frac{1}{2}$	$\alpha,\beta,\gamma \in \mathbb{Z}$	$\gcd(2\alpha,2\beta,2\gamma)=1$ or 2

One important invariant is the discriminant:

	Δ
A_β	$-4\det A_\beta$
(a,b,c)	b^2-4ac
(a,b',c)	$\sum_{cyc}(a^2-2ab')$
(α,β,γ)	$-4\sum_{cyc}\alpha\beta$

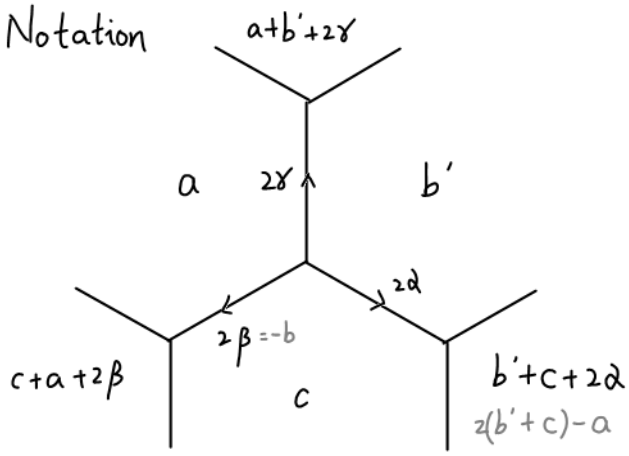


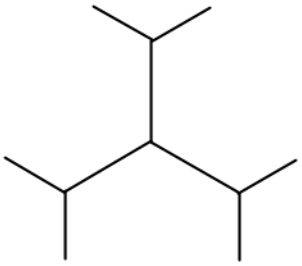
Geometrical explanation for the discriminant

Moving along the river

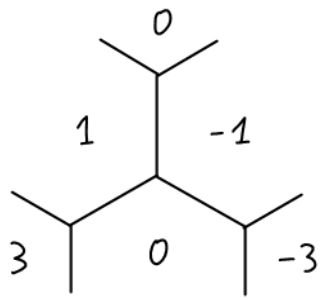
Here I collect some examples, so I can see in the future what's happening in the indefinite case.
You should definitely try this one:
https://xenon.stanford.edu/~hwatheod/topograph/topograph.html

Notation

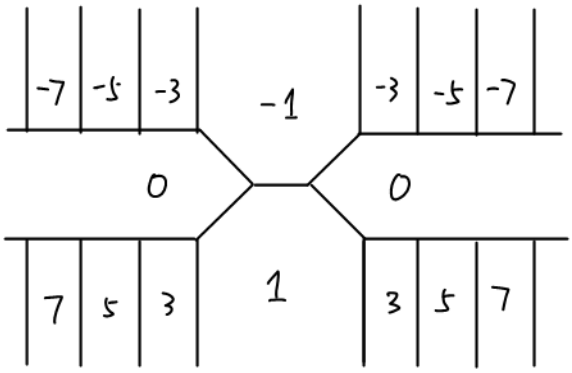


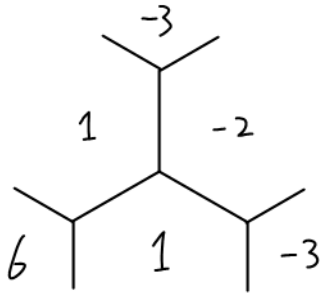


E.g.

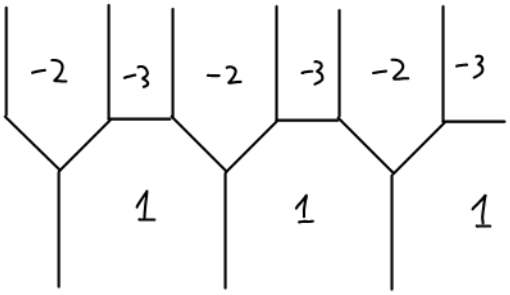


$x^2 - 2xy$

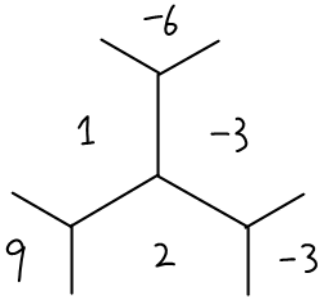




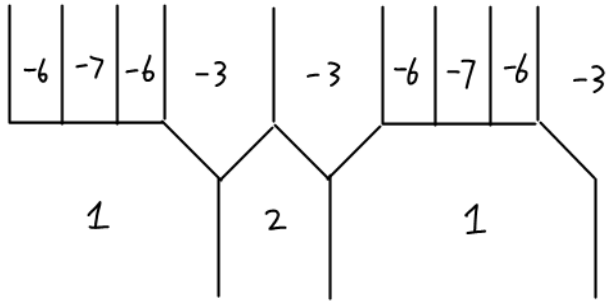
$x^2 - 4xy + y^2$



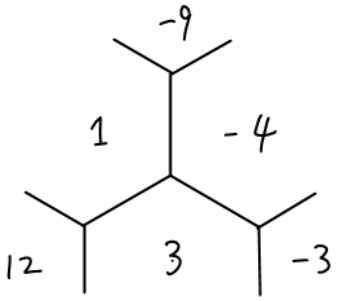
period: 3



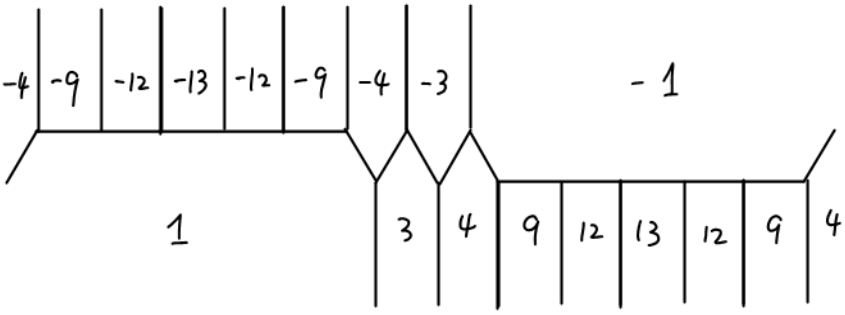
$x^2 - 6xy + 2y^2$



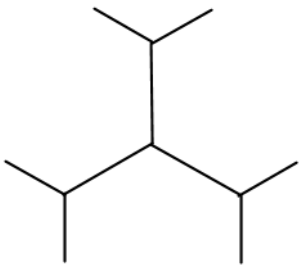
period: 7



$x^2 - 8xy + 3y^2$



period: 20



walk along your river!